Streamlined Failure Localization Method and Application to Network Health Monitoring

Camille Delezoide, *Member, IEEE*, Petros Ramantanis, *Member, IEEE*, and Patricia Layec, *Member, IEEE*

***Abstract*—We propose a reliable and easy-to-deploy method for localizing failures in optical networks with low added complexity. It essentially relies on the aggregation of raw monitoring data from coherent transponders; data readily accessible through streaming telemetry in modern optical networks. We first verify the reliability of our algorithm through extensive simulations over a European topology under realistic conditions. Then, we apply it to field data from a backbone network carrying live traffic. Here, we show how failure localization allows to generate strategic network health insights regarding link failure probabilities and mean time to repair. We notably underline the involvement of Q-drops in strong and localized deviations of both parameters.**

***Index Terms*—Failure localization, availability, Q-drops, field trial, monitoring, optical networks.**

I. Introduction

AVAILABILITY is one of the three key performances of any communication network, with capacity and latency. Because of their paramount role in modern communications, optical networks are particularly expected to be highly available, typically above five-nines. Yet, reality on the ground may be largely below such expectations. Indeed, Ghobadi et al. recently showed that availability can vary over more than three orders of magnitude in a large backbone network [1]. More, we demonstrated in [2], [3] that the transient events termed *Q-drops* in [1] are often short-lived packet losses that remain undetected at upper layers. Considering the added risk of malicious attacks [4] and increasing frequency of network-impacting natural disasters [5], solutions to maintain if not improve availability are more than ever needed.

Introducing network margins [6] and/or redundancies such as optical protection are popular techniques to boost availability. Yet, both are highly costly in terms of effective spectral efficiency [7]. Thus, the current challenge is not to improve availability, but to research more efficient ways to do it. In this context, directly improving availability by reducing the mean-time-to-repair (MTTR) is appealing. The MTTR can be decomposed into the time it takes to localize the failure, and the time required to fix it. Although the latter contribution is difficult to improve without increasing costs, failure localization has the potential to be accurate, cost-efficient and almost instantaneous. Coupled to proactive maintenance, it may also effectively prevent failures and further improve availability.

Active failure localization methods require dedicated monitoring equipment: optical time domain reflectometer (OTDR) [8], variations of probe lightpaths [9], [10], [11], [12], [13], [14] and low-cost optical spectral analyzers (OSA) [15]. In contrast, recently introduced *passive* methods simply leverage default network equipment and streaming telemetry, thus promising lower deployment costs. For pinpointing fiber cuts, they could be coupled to receiver-based power profile estimation techniques [16] or OTDR. Proposed passive methods employ network kriging (NK) [17], [18], statistical machine learning (SML) [19], Gaussian noise (GN) model-based parameter estimation [20], knowledge graphs (KG) [21], and belief networks (BN) [22].

Despite its practical importance, a generally overlooked limitation of all failure localization methods is ambiguity, i.e. the coexistence of multiple failure scenarios explaining a given set of observations, including the possibility of false alarms. Ambiguous cases may only be partially solved by some decision mechanism, e.g. by determining which scenario is the likeliest. However, there is then no guarantee to provide the true failure location, hence the need for field technicians to further investigate. Thus, any automated failure localization - including our own - may merely expedite the troubleshooting process by reducing required verifications, but cannot systematically solve the failure location.

With generally scarcer monitoring endpoints, passive methods are potentially more impacted by ambiguity than active methods. This is a counterpart of their lower deployment cost. Passive methods thus require particular attention to this issue. The inherent risk of localization errors imputable to ambiguity underlines the necessity of developing solutions with *interpretable* outputs. In particular, this does not seem to be the case yet for ML-based solutions proposed in [19], [21], [22]. Due to training requirements and algorithmic complexity, the same solutions seem to presently lack the *flexibility* required in modern optical networks, where reconfigurations are increasingly frequent. This contrasts with the lightweight NK-based method where the core input - the routing matrix - can easily be updated whenever the network is reconfigured. Its limitation however is its inability to handle ambiguous cases without resorting to probe lightpaths. Another related aspect is the ability to handle multiple simultaneous failures, which are not considered in [19],

Manuscript received 19 January 2023; revised 21 April 2023; accepted 16 May 2023. Date of publication 8 June 2023; date of current version 2 October 2023. This work was supported by AINET-ANTILLAS. *(Corresponding author: Camille Delezoide.)*

The authors are with the Optical System and Device Research Lab, Nokia Bell Labs, Massy 91300, France (e-mail: camille.delezoide@nokia-bell-labs.com; petros.ramantanis@nokia-bell-labs.com; patricia.layec@nokia-bell-labs.com).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/JLT.2023.3282447.

Digital Object Identifier 10.1109/JLT.2023.3282447

TABLE I
COMPARISON BETWEEN VARIOUS PASSIVE SOLUTIONS FOR FAILURE LOCALIZATION

Solution	**Deployability**	**Interpretability**	**Flexibility**
NK [17]	Low-Med	High	High
SML [19]	Med-High	Low	Low
GN-based [20]	High	Low-Med	High
KG [21]	Med	Low	Low
BN [22]	Med-High	Low	Low
SFL (this paper)	High	High	High

[20]. After careful analysis of various technical aspects including the comments above, we propose a qualitative comparison with prior methods in Table I, where *deployability* broadly relates to ease of deployment, which accounts for overall complexity and deployment cost.

In this article, we propose a passive, streamlined failure localization (SFL) method combining the simplicity of the NK approach with statistical inference to partially disambiguate complex failure scenarios. We particularly focused on achieving full support of multiple failures and delivering highly interpretable results. In the following, we first propose a simplified mathematical formalism and method which, upon any failure, efficiently converts raw monitoring data and topological information into a list of possible failure scenarios. After discussing how failure scenarios may be ranked according to their likelihoods, we test our method by simulating single and double failures in a European backbone topology. Finally, revisiting our previous work [2], we apply SFL to study the failures of an active backbone optical network monitored during a field trial between April 2020 and March 2021. We particularly leverage failure localization to generate statistical insights on which portions of the studied network are more susceptible to failures, including Q-drops.

II. LOCALIZATION METHOD

Throughout the article, we define links as interconnections between two nodes. In any transparent optical network, a failure such as a fiber cut necessarily impacts the quality of transmission (QoT) for all lightpaths going through the failed link. Thus, by generating QoT-based alarms at the lightpath level and aggregating them at the network level, it is possible to localize a failure without any additional monitoring equipment. Yet, this ability fundamentally relies on lightpath diversity. In the following, we introduce the notion of segments when considering the localization problem. Then, we introduce the mathematical formalism used throughout our algorithm. Finally, we describe the three steps of our localization algorithm: segments exclusion, listing possible failure scenarios and ranking scenarios by likelihoods.

A. Pre-Processing: From Links to Segments

When lightpath diversity is low, it can be impossible to localize a failed link simply by processing receiver-based alarms. For instance, if two connected links are solely traversed by the same lightpath, it is possible to determine that one of the two links failed, but impossible to determine which. In our method, we circumvent this limitation by regrouping links whose failures cannot be distinguished into groups that we define as *segments*. We can show that links belonging to the same segment are the links traversed by exactly the same lightpaths. We use this property to identify the segments at the output of the routing and wavelength assignment (RWA) algorithm, then to generate the routing matrix $\mathbf{R}$ mapping the lightpaths on the segments. The failure localization algorithm we propose is thus limited to the segment level. Yet, we will see that for the most common network configurations, all links failures can be distinguished making the segment equivalent to the link.

B. Mathematical Formalism

We define the alarm vector $\mathbf{A}$ as a synthesis of all modifications of individual lightpaths alarms, all synchronously polled every period T. It is a logical column vector with N rows, where N is the number of active lightpaths. At instant t, any component $\mathrm{A_k(t)}$ of $\mathbf{A}(\mathrm{t})$ is equal to 1 if the alarm from lightpath k went from OFF to ON state between $t-T$ and t, and is equal to 0 otherwise. To describe the changes of segments statuses, we define the failure vector $\mathbf{F}$ representing a failure scenario as a logical column vector with M rows, M being the number of segments. At instant t, any component $\mathrm{F_j(t)}$ of $\mathbf{F}(\mathrm{t})$ is equal to 1 if segment j failed between $t-T$ and t, and is equal to 0 otherwise. The routing matrix $\mathbf{R}$ is a N-by-M logical matrix. The component $\mathrm{R_{k,j}}$ of $\mathbf{R}$ is equal to 1 if lightpath k goes through segment j, $\mathrm{R_{k,j}} = 0$ otherwise. The columns of the routing matrix represents the segments (S) while the rows represent the lightpaths (LP). Assuming that any segment failure directly translates into alarms rises for the lightpaths going through the failed segment, we verify:

$$\mathbf{A} = \mathbf{R} \cdot \mathbf{F} \tag{1}$$

The equation above describes that given the routing matrix $\mathbf{R}$, a failure scenario $\mathbf{F}$ leads to the generation of a non-zero alarm vector $\mathbf{A}$. Ultimately, the goal of our localization algorithm is to invert this equation, i.e. to determine the correct value of $\mathbf{F}$ based on the knowledge of $\mathbf{R}$ and the observation of $\mathbf{A}$. Mathematically-speaking, this is equivalent to determining $\mathbf{R^+}$, the left pseudoinverse of the routing matrix $\mathbf{R}$ verifying:

$$\mathbf{F} = \mathbf{R}^{+}\mathbf{R}\mathbf{F} = \mathbf{R}^{+}\mathbf{A}$$

The necessary conditions for the existence of the pseudoinverse $\mathbf{R^+}$ are generally not met, which relates to the ambiguity issue discussed in Section I. Indeed, multiple failure scenarios $\mathbf{F}$ may verify (1), including scenarios with multiple simultaneous failures. The algorithm described below first finds all scenarios verifying (1). If $\mathbf{R^+}$ exists, the failure localization is directly given by the sole output scenario. In ambiguous cases, the correct scenario is among the various solutions found. Those are then ranked according to their estimated likelihoods in order to efficiently define priorities when deploying resources to isolate and repair the failure(s).

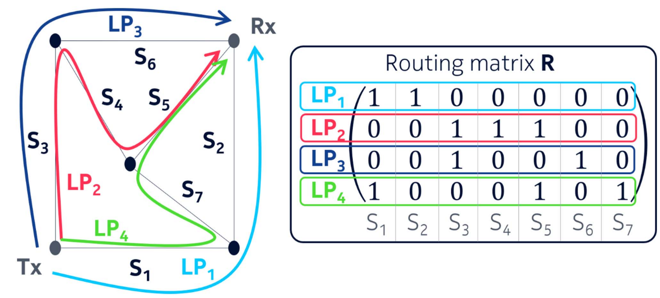

Fig. 1. Example topology and associated routing matrix.

C. First Step of Localization: Segments Exclusion

This first step consists in excluding from the list of potentially failed segments those that cannot be involved in the observation of **A** given the routing matrix **R**. Although this step is not strictly necessary, it drastically improves computational efficiency by reducing the number of possibilities to explore. We achieve this by comparing **A** with columns of **R**. In particular, we can show that if $\mathrm{A_k} = 0$, it is logically impossible for any segment j traversed by lightpath k to be faulty. As an example, let us consider the alarm $\mathbf{A} = (1\ 0\ 0\ 1)$ in the example topology described in Fig. 1. Since $\mathrm{A_2} = \mathrm{A_3} = 0$, any segment j verifying $\mathrm{R_{2,j}} = 1$ or $\mathrm{R_{3,j}} = 1$ or both cannot be in failed state. This concerns segments 3, 4, 5 and 6, which can thus be excluded from the search.

D. Second Step of Localization: Listing Failure Scenarios

We discussed that given **R** and depending on the observation **A**, there can be multiple vectors **F** verifying (1), i.e. multiple failure scenarios leading to the observation of **A**. In each scenario, one or multiple segments may be simultaneously involved. We obtain an exhaustive list of all scenarios given **R** and **A** using a brute-force method. This method consists in constructing all possible **F** vectors involving the segments that have not been excluded in the previous step. Then, the vectors found to verify (1) represent all possible failure scenarios. An alternative to brute force would be to apply additional logical selection rules to further reduce the number of possibilities during the search, thus making the algorithm converge faster.

In our example with the alarm $\mathbf{A} = (1\ 0\ 0\ 1)^\intercal$, $^\intercal$ standing for vector transpose, this provides the exhaustive list of failure scenarios involving the non-excluded segments 1, 2 and 7:

- $\mathbf{F_a} = (1\ 0\ 0\ 0\ 0\ 0\ 0)^\intercal$: a single failure of segment 1 is a possible failure scenario for $\mathbf{A} = (1\ 0\ 0\ 1)^\intercal$.
- $\mathbf{F_b} = (0\ 1\ 0\ 0\ 0\ 0\ 1)^\intercal$: failures of segments 2 and 7
- $\mathbf{F_c} = (1\ 1\ 0\ 0\ 0\ 0\ 0)^\intercal$: failures of segments 1 and 2
- $\mathbf{F_d} = (1\ 0\ 0\ 0\ 0\ 0\ 1)^\intercal$: failures of segments 1 and 7.
- $\mathbf{F_e} = (1\ 1\ 0\ 0\ 0\ 0\ 1)^\intercal$: failures of segments 1, 2 and 7.

Since alarm states are logical values, one failed segment can partially or fully mask the failure of another segment. This mechanism is involved in the three last failure scenarios.

E. Third Step of Localization: Ranking Scenarios

This final step consists in ranking failure scenarios according to their likelihoods. It relies on estimating the failure probabilities of involved links, i.e. the probabilities that the links will fail in any time window of duration T corresponding to the refresh period of the alarms. This is routinely done in optical communications to evaluate the availabilites of lightpaths at the design stage. They can also be made after deployment based on field observations. Then, segment failure probabilities are calculated as the sum of the failure probabilities of their links. Finally, a scenario's likelihood is the probability that all involved segments simultaneously fail.

Typically, segment failures are independent so that the scenario's likelihood is the product of the failure probabilities for involved segments. However, some events can lead to the simultaneous failures of multiple segments. Those can occur on relatively small scales, e.g. the failure of a node connecting multiple links. They can also occur on much broader scales in the case of natural disasters [23]. The probabilities of such events can be much larger than the probability that the same segments simultaneously fail for independent reasons. Thus, an optimal estimation of failure scenarios likelihoods would require accounting for the probabilities of joint failures. This will be the object of further investigations. In the following, we will assume for simplicity that all segment failures are statistically independent.

In our example, let us assume that the failure probability for all segments is $p_0 < 1$. We further assume that the segments are independent, i.e. that any two segments cannot fail from the same event. Under those assumptions, we obtain that the likeliest scenario is $\mathbf{F_a}$, with a probability $\mathrm{p}(\mathbf{F_a}) = p_0$. Then, we have three equally-probable scenarios with $\mathrm{p}(\mathbf{F_b}) = \mathrm{p}(\mathbf{F_c}) = \mathrm{p}(\mathbf{F_d}) = p_0^2$. Finally, the less likely scenario is $\mathbf{F_e}$ with $\mathrm{p}(\mathbf{F_e}) = p_0^3$. As numerical example, $p_0 = 10^{-2}$ makes $\mathbf{F_a}$ 100 times likelier than $\mathbf{F_b}$, $\mathbf{F_b}$ or $\mathbf{F_d}$, and 10000 times likelier than $\mathbf{F_e}$. Thus, assuming that $\mathbf{F_a}$ is the correct scenario would only be true in 97.1% of cases. Although acceptable here, this accuracy would significantly decrease with larger failure probabilities, or simply in cases where all possible scenarios have comparable likelihoods. The latter cases will typically require more operational efforts to solve. Yet, this challenge stems from the routing matrix and the failure itself, and not from the algorithm.

F. Adaptation to Network Reconfigurations

Since the localization method dominantly relies on the routing matrix **R**, this input must be updated whenever existing lightpaths are rerouted or new lightpaths are introduced. This entails to renew the pre-processing step, which may result in a different number of segments. In such case, the failure vector **F** must be resized accordingly, so does the alarm vector **A** to account for newly introduced lightpaths, if any.

III. Application to a Full-Scale Network

A. Simulation Method

To test the localization algorithm in realistic conditions, we select the European backbone topology depicted in Fig. 2 with 28 nodes connected by a total of 41 bidirectional links. The goal is to test the algorithm for a large variety of network configurations

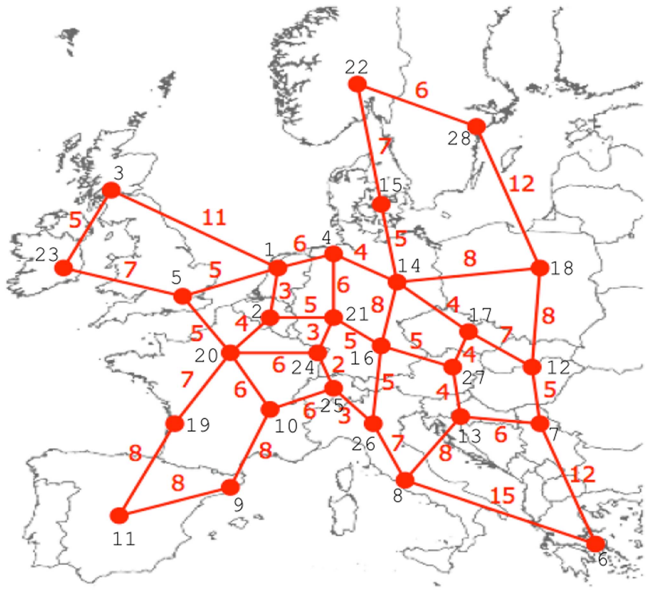

Fig. 2. Topology for the simulation of the localization algorithm.

through multiple simulation runs. In each run, we start by randomly selecting $\mathrm{N_{LP}}$ combinations of ingress and egress nodes. Then, we use a Dijkstra RWA algorithm with first fit allocation to determine the corresponding lightpaths. After pre-processing, we obtain the routing matrix **R** mapping the $\mathrm{N_{LP}}$ lightpaths on $\mathrm{N_{seg}}$ network segments. Finally, we generate failure vectors **F** and compare them with the localization algorithm's outputs. In the following, we discuss the results obtained after making 10 consecutive runs with random lightpaths configurations for all numbers of lightpaths $\mathrm{N_{LP}}$ between 1 and 340, hence a total of 3400 simulation runs. Note that $\mathrm{N_{LP}} = 340$ roughly corresponds to the saturation of the network in our simulations.

B. Outcome of Failure Scenario Listing

We focus here on the number of failure scenarios identified through the localization algorithm. Naturally, the ideal case is when the algorithm identifies a single scenario. Indeed, this implies a direct localization without the need for any assumption on failure probabilities. For each run, we test all possible single and double failures, i.e. $\mathrm{N_{SF}}$ and $\mathrm{N_{DF}}$ respectively, verifying:

$$N_{SF} = N_{seg} \qquad N_{DF} = \frac{N_{seg}\left(N_{seg}-1\right)}{2}$$

For all simulated failure vectors, we simply count the number of failure scenarios verifying (1). For plotting purposes, we summarize the results as case ratios, separating single and double failures. For instance, the unambiguous case ratio is the number of times a simulated single failure led to the identification of unique failure scenario divided by the number of simulated single failures, i.e. $\mathrm{N_{SF}}$. In Fig. 3, we plot the averages over the 10 runs found for the case ratios as functions of the number of lightpaths $\mathrm{N_{LP}}$.

As can be expected, the number of possible failure scenarios significantly depends on $\mathrm{N_{LP}}$. When $\mathrm{N_{LP}}$ is small i.e. below 10, the relatively low level of ambiguity in the failure localization hides the fact that although faulty segments can easily be identified, segments are composed of multiple links due to the low lightpath diversity. Here, the effective localization ambiguity lies in that it is formally impossible to determine which link of the segment is faulty. For $\mathrm{N_{LP}}$ between 10 and 50, the ambiguity measured in terms of amounts of possible failure scenarios for each simulated failure is at its highest. This is a result of an exponential increase of potential failure combinations while the lightpath diversity remains low, due to the small number of lightpaths compared to the number of links. For $\mathrm{N_{LP}}$ above 50, the ambiguity decreases. Additionally, the increasing lightpath diversity quickly leads to a situation where segments are almost always equivalent to links, with rare exceptions on outer parts of the network. Consequently, we show that this most common situation for mature networks is also the most favorable for direct failure localization at the link level. Near full network load i.e. for $\mathrm{N_{LP}} > 300$, more than 90% of single failures can be localized without ambiguity, and more than 99% of failure cases will have only two or less failure scenarios. Overall, the algorithm shows a slightly decreased performance for double failures compared to single failures, due to the increased complexity of the generated alarms. Still, near full network load roughly 85% of double failure cases can be localized without ambiguity, and similarly to single failures, more than 99% of cases have only two or less failure scenarios.

C. Ranking Scenarios by Likelihood

In the typically rare cases where the output of the second step of the localization is ambiguous, the third and final step of the localization algorithm ranks the failure scenarios according to their estimated probabilities of occurrence. This is meant for maintenance operations to check failures origins by decreasing likelihood based on the insights provided by the algorithm. Indeed, this is the most efficient maintenance strategy to reduce the MTTR, which is the core rationale for automated failure localization.

In order to evaluate the probabilities of the failure scenarios in our simulations, we make the typical assumption that failure probabilities significantly vary from one link to another, for instance depending whether the fiber cable is aerial or underground, and also depending on its length. Based on field data where availability was found to range by more than three orders of magnitude across a large backbone network [1], we considered a uniform distribution for the log of the failure probabilities so that simulated values are distributed between 10^{-1} and 10^{-5}. The model for estimating failure probabilities could be made more complex, for instance by accounting for distinct probabilities for unidirectional and bidirectional link failures, or node failures impacting two or more links at the same time. Yet, we assumed here for simplicity that link failures were independent so that the failure probability of a given segment is equal to the sum of the failure probabilities of its links, and that the probability of a double failure is always the product of the two failure probabilities.

In Fig. 4, we plot the ratios of ambiguous cases for which the actual - i.e. simulated - failure scenario is ranked first, second, or third and beyond in the list of possible failure scenarios. As

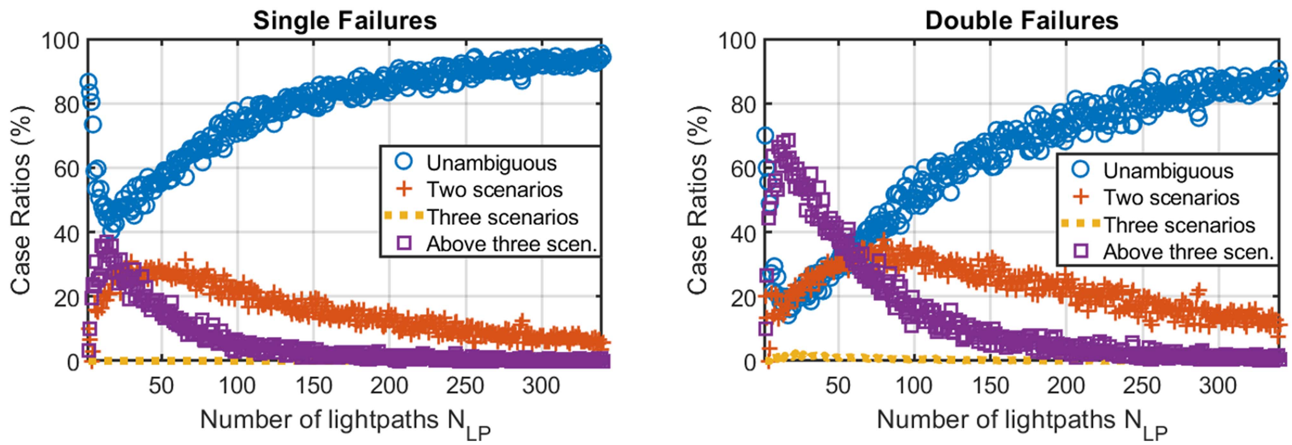

Fig. 3. Case ratios as percentages of cases found to correspond to the plot's legend considering all simulated failures. The "unambiguous" plot corresponds to the cases where only one possible failure scenario is found, for which the localization is direct.

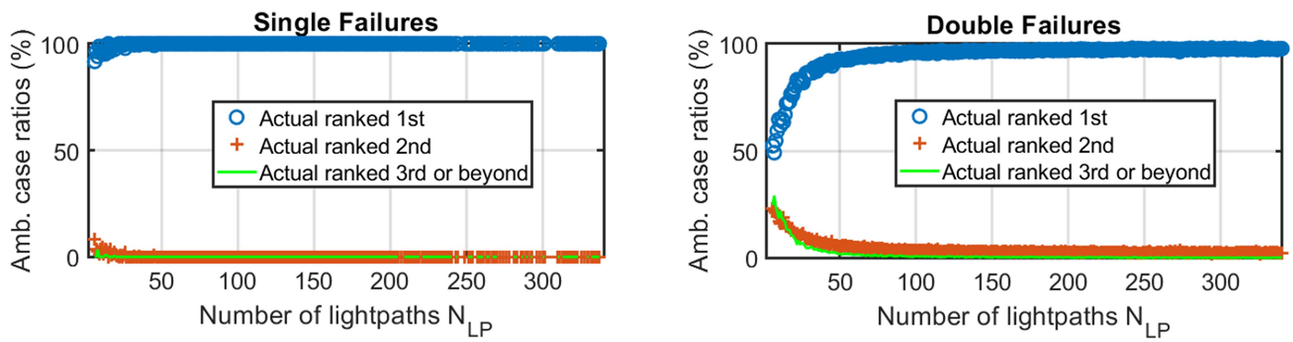

Fig. 4. Case ratios as percentages of ambiguous cases found to correspond to the plot's legend considering all simulated failures. Values are averaged over 10 simulation runs, each with a unique random network configuration. The "actual" failure scenario is the simulated failure scenario.

previously, these ratios are averages over 10 simulation runs. For single failures, we obtain that for $N_{LP} > 50$, the most likely scenario is always the simulated one. For double failures however, we observe that the number of lightpaths has a much larger impact on the results. In particular, near full network load there is a residual percentage of ambiguous cases, below 5%, for which the actual failure scenario is only ranked in second position. After further investigation, we found that this occurs when the alarm vector from a single failure incidentally matches that corresponding to the simulated double failure, which is typically less likely. Here as for any other ambiguous case where the correct scenario is not the likeliest, the failure scenarios would have to be examined by a third-party method, for example sequentially according to their ranking. This could be made automatically by cross-checking the output of the localization algorithm with the span attenuations reported by the inline amplifiers, or error messages. Yet, our results show that in the vast majority of cases, the simulated failure scenario is ranked first. This means that even for ambiguous cases, the most likely scenario is a very robust indicator of the actual failure scenario. Finally, we can remark that if available, directly cross checking the failure scenarios found by our algorithm to such independent sources of information would automatically lead to more efficient and robust localization.

IV. Application to Field Trial Data

In this section, we leverage monitoring data from a field trial to test the operational deployment of our failure localization algorithm. Within a national backbone optical network carrying live traffic, we installed Nokia PSI-2 T transponders at two distant locations to build a total of 4 distinct lightpaths, each delivering a capacity of 100 Gb/s with PDM-QPSK modulation format. The topology of the trial is similar to that depicted in Fig. 1, with 7 segments. We polled the values of pre and post-FEC BER for the four lightpaths every second for 10 months. We limit the study to a single direction.

During the trial, 22 failure events concerning the monitored lightpaths have been reported through the network management suite (NMS), strictly applying the ITU-T standards in force [24]. Such events have typically been identified as resulting from fiber cuts. In parallel, from the monitored post-FEC BER data during the same period, we count a total of 160 events where the post-FEC BER of at least one lightpath has been above 0 for at least one monitoring period, i.e. 160 packet loss events. We attribute the major discrepancy between the number of failures reported by the NMS and that of packet loss events to how failures are defined in standards. The fact that 138 short-lived packet loss events were never reported as failures by the NMS points out

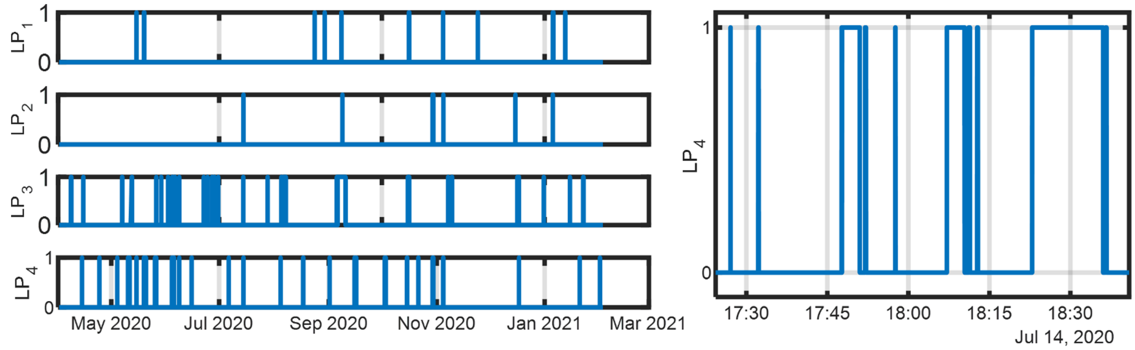

Fig. 5. Failure status derived from the observation of the monitored post-FEC BER for all lightpaths during the 10 months of the field trial and zoom on a sequence of short busts of post-FEC errors observed for LP_4.

the necessity to directly process raw monitoring data rather than high-level alarm messages when it comes to accurately survey availability levels. In Fig. 5, we summarize the observations of lightpaths failures during the entire field trial. We also zoom in on LP_4 as example of a sequence of short bursts of post-FEC errors with varying durations.

To analyze the locations of all 160 outages regardless of their durations, we defined the alarm vector $\mathbf{A}$ described in Section II-B as a 4×1 logical vector describing where $\mathrm{A_k(t)} = 1$ only if lightpath k goes from a zero to a non-zero post-FEC BER at time t. In our previous study on the same raw data [2], we counted a total of 130 failures, 4 being directly identified as simultaneous failures of two segments, and 14 being ambiguous cases. Here, we report slightly different results for two reasons. First, our former study investigated failure locations directly from post-FEC BER statuses while here, the proposed algorithm processes variations of these statuses. This results in less ambiguity since any alarm status resulting from two independent but consecutive failures - formerly interpreted as one ambiguous period - is now treated as two successive and unambiguous failure events. Secondly, we leverage here the native one-second resolution of the dataset whereas our prior study evaluated failure statuses with one-minute resolution. This results here in a larger count of failures as some of the failures were less than one minute apart. The finer time resolution also allows to find out that previously reported double failures were actually distinct single failures occurring within the same one-minute window.

Out of the 160 failure events observed, 145 events unambiguously point to a single failure scenario. For all 15 remaining events, there are five possible failure scenarios, only one of which being a single segment failure. This is segment 1 for the $(1\ 0\ 0\ 1)^\mathsf{T}$ alarm, and segment 5 for the $(0\ 1\ 0\ 1)^\mathsf{T}$ alarm. In the following, we make the hypothesis that for the 15 ambiguous cases, the actual failure scenario is the single segment failure. We later use the results obtained to check the consistency of this hypothesis and its potential impact on the results. As summary of the network health diagnosis allowed by failure localization, we display in Table II for each segment and for the entire duration of the trial the number of observed failures, the failure probability defined here as the probability that the segment will fail in any given second, the failure probability divided by the length of the segment, and finally the mean-time-to-repair i.e. the average duration of the observed segment failures.

TABLE II
FIELD TRIAL FAILURE STATISTICS

Segment	Failures	P_{fail}	P_{fail}/km	MTTR (h)
1	5	1.88e-7	1.25e-9	1.17
2	9	3.39e-7	1.22e-9	2.43
3	0	0	0	0
4	6	2.26e-7	1.59e-9	6.86
5	10	3.76e-7	1.30e-9	1.04
6	67	2.52e-6	2.86e-9	2.35
7	64	2.41e-6	4.46e-8	0.16

During the 10 months of the trial, only segment 3 did not fail. This can be explained by the fact that this segment is a very short underground cable connecting two network exchanges. For all other segments, the highest failure probability observed is 2.52e-6 for segment 6. Considering this probability, the likelihood of a double failure for independent segments would be roughly six orders of magnitude lower than that of a single failure. This supports the hypothesis made for the trial that for ambiguous cases, the correct failure scenario is the single failure one. We also point out that the extremely rare situations where double failures could be mistaken for single failures would not significantly modify the displayed statistics. When comparing the failure probabilities segment by segment, a first important observation leveraging the localization is that the failure probability per km of fiber is almost the same for segments 1, 2, 4 and 5 with an average at 1.34e-9 and a standard deviation of 0.17e-9. These segments are similar in the sense that they are mixes of aerial and underground fiber spans crossing mostly urban areas. In contrast, segment 6 is longer than all other segments and mostly composed of aerial fiber spans crossing rural areas. This seemed to result in a higher failure probability per km, at 2.86e-9. The highest failure probability per km was found to be 4.68e-8 for segment 7, more than 30 times higher than for segment 2 for instance. With a failure probability similar to segment 6 while being much shorter, segment 7 appeared as an anomaly. Comparing both segments, we finally observed that the anomalously high failure probability of segment 7 was mainly due to the prevalence of the Q-drop phenomenon for this specific segment. This appears in the significantly lower MTTR for segment 7 than for any other segment.

V. Conclusion

We proposed an algorithm to localize the origin of optical layer failures solely from the analysis of monitoring data collected from coherent transponders at the network scale. We tested the algorithm through extensive simulations on a complex network topology. We demonstrated that for typical network configurations, more than 90% of failures can be directly localized without ambiguity, and more than 99% of failures are correctly localized after considering the likelihoods of possible failure scenarios for ambiguous cases. Finally, we deployed this algorithm in a backbone network carrying live traffic to study occurrences of post-FEC errors. We showed how our localization method can generate individual failure statistics for network segments, and how strategic insights on network health can be deduced from it.

Acknowledgment

The authors warmly thank Fabien Boitier, Emmanuel Séve, Lluis Gifre and Sylvain Chenard for fruitful discussions, and our silent partners for granting us access to their network, and for their multiple contributions that made the field trial possible.

References

[1] G. Monia et al., "Optical layer failures in a large backbone," in *Proc. Internet Meas. Conf.*, 2016, pp. 461–467. doi: 10.1145/2987443.2987483.

[2] C. Delezoide, P. Ramantanis, L. Gifre, F. Boitier, and P. Layec, "Field trial of failure localization in a backbone optical network," in *Proc. Eur. Conf. Opt. Commun.*, 2021, pp. 1–4.

[3] C. Delezoide, P. Ramantanis, and P. Layec, "Investigating Q-drops and their probable causes," in *Proc. Eur. Conf. Opt. Commun.*, 2022, pp. 1–4.

[4] N. Skorin-Kapov, M. Furdek, S. Zsigmond, and L. Wosinska, "Physical-layer security in evolving optical networks," *IEEE Commun. Mag.*, vol. 54, no. 8, pp. 110–117, Aug. 2016.

[5] S. Banholzer, J. Kossin, and S. Donner, "The Impact of Climate Change on Natural Disasters," in *Reducing Disaster: Early Warning Systems for Climate Change*. Berlin, Germany: Springer, 2014, pp. 21–49.

[6] C. Delezoide, P. Ramantanis, and P. Layec, "Leveraging field data for the joint optimization of capacity and availability in low-margin optical networks," *J. Lightw. Technol.*, vol. 38, no. 24, pp. 6709–6718, Dec. 2020.

[7] C. Delezoide, P. Ramantanis, and P. Layec, "Machine Learning and Data Science for Low-Margin Optical Networks," in *Machine Learning for Future Fiber-Optic Communication Systems*, A. P. T. Lau and F. N. Khan, Eds. Orlando, FL, USA: Academic Press, 2022, ch. 9, pp. 281–315.

[8] I. Sankawa, S. Furukawa, Y. Koyamada, and H. Izumita, "Fault location technique for in-service branched optical fiber networks," *IEEE Photon. Technol. Lett.*, vol. 2, no. 10, pp. 766–768, Oct. 1990.

[9] C. Mas, I. Tomkos, and O. Tonguz, "Failure location algorithm for transparent optical networks," *IEEE J. Sel. Areas Commun.*, vol. 23, no. 8, pp. 1508–1519, Aug. 2005.

[10] Y. Wen, V. W. S. Chan, and Lizhong Zheng, "Efficient fault-diagnosis algorithms for all-optical WDM networks with probabilistic link failures," *J. Lightw. Technol.*, vol. 23, no. 10, pp. 3358–3371, Oct. 2005.

[11] S. S. Ahuja, S. Ramasubramanian, and M. Krunz, "SRLG failure localization in all-optical networks using monitoring cycles and paths," in *Proc. IEEE INFOCOM*, 2008, pp. 700–708.

[12] J. Tapolcai, B. Wu, and P.-H. Ho, "On monitoring and failure localization in mesh all-optical networks," in *Proc. IEEE INFOCOM*, 2009, pp. 1008–1016, doi: 10.1109/INFCOM.2009.5062012.

[13] S. Stanic and S. Subramaniam, "Fault Localization in All-Optical Networks With User and Supervisory Lightpaths," in *Proc. Int. Conf. Commun.*, 2011, pp. 1–6.

[14] Y. Xuan, Y. Shen, N. P. Nguyen, and M. T. Thai, "Efficient multi-link failure localization schemes in all-optical networks," *IEEE Trans. Commun.*, vol. 61, no. 3, pp. 1144–1151, Mar. 2013.

[15] A. P. Vela et al., "Soft failure localization during commissioning testing and lightpath operation," *J. Opt. Commun. Netw.*, vol. 10, no. 1, pp. A27–A36, Jan. 2018.

[16] A. May, F. Boitier, A. Courilleau, B. A. Ayoubi, and P. Layec, "Demonstration of enhanced power losses characterization in optical networks," in *Proc. Opt. Fiber Commun. Conf.*, 2022, Paper Th1C-6, doi: 10.1364/OFC.2022.Th1C.6.

[17] K. Christodoulopoulos, N. Sambo, and E. Varvarigos, "Exploiting network kriging for fault localization," in *Proc. Opt. Fiber Commun. Conf.*, 2016, Paper W1B.5.

[18] C. Delezoide et al., "Pre-emptive detection and localization of failures towards marginless operations of optical networks," in *Proc. 20th Int. Conf. Transparent Opt. Netw.*, 2018, pp. 1–4.

[19] T. Panayiotou, S. P. Chatzis, and G. Ellinas, "Leveraging statistical machine learning to address failure localization in optical networks," *J. Opt. Commun. Netw.*, vol. 10, no. 3, 2018, Art. no. 162.

[20] S. Barzegar, "Soft-failure localization and device working parameters estimation in disaggregated scenarios," in *Proc. Opt. Fiber Commun. Conf.*, 2020, Paper Th1F.2, doi: 10.1364/OFC.2020.Th1F.2.

[21] Z. Li, Y. Zhao, Y. Li, S. Rahman, X. Yu, and J. Zhang, "Demonstration of fault localization in optical networks based on knowledge graph and graph neural network," in *Proc. Opt. Fiber Commun. Conf.*, 2020, Paper Th1F.5, doi: 10.1364/OFC.2020.Th1F.5.

[22] R. Liang, F. Liu, and J. Liu, "A belief network reasoning framework for fault localization in communication networks," *Sensors*, vol. 20, no. 23, Dec. 2020, Art. no. 6950.

[23] M. Shiraiwa et al., "Experimental demonstration of disaggregated emergency optical system for quick disaster recovery," *J. Lightw. Technol.*, vol. 36, no. 15, pp. 3083–3096, Aug. 2018.

[24] *Error Performance of an International Digital Connection Operating At a Bit Rate Below the Primary Rate and Forming Part of an Integrated Services Digital Network*, ITU-T REC. G.821, International Telecommunications Union, Geneva, Switzerland, Dec. 2002.